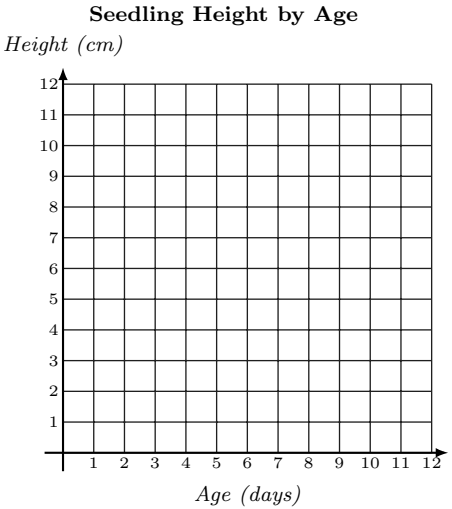
Artifact 18 – Scatterplots and Association

Driving Question: How do you build a scatterplot from a table, and what do the points tell you about how two variables are related?

Example 1

The table shows the age, in days, of twelve bean seedlings and the height of each, in centimeters. Plot the data.

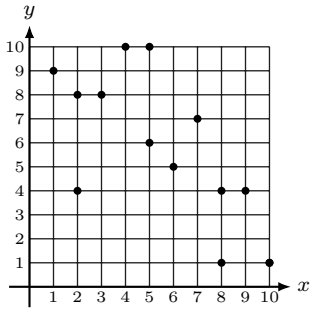


Age (days)	Height (cm)
1	3
2	4
3	2
4	6
5	8
6	8
7	5
8	11
9	9
10	8
11	7
12	10

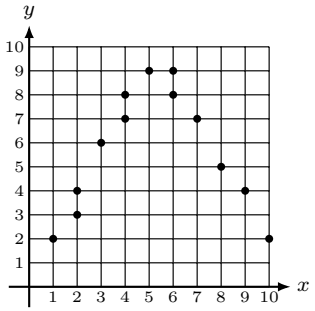
Describe the association shown by the data.

Example 2

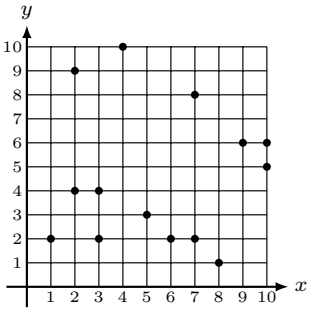
Each scatterplot shows a different set of data. Describe the association in each.



x: age of a tire (years)
y: tread depth (mm)



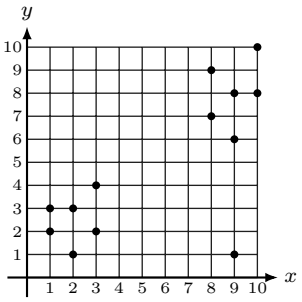
x: lemonade price (dollars)
y: daily profit (tens of dollars)



x: letters in first name
y: game score

Example 3

The scatterplot shows the length of each of thirteen hikes and the water the hiker used.



x: hike length (miles)
y: water used (cups)

Describe the association shown.

Identify any cluster or outlier. Describe each in the context of the hikes.

Vocabulary

bivariate data data that records two variables for each person or thing measured, written as ordered pairs (x, y)

scatterplot a graph of plotted points that shows the relationship between a bivariate set of data

association the overall pattern between the two variables in a scatterplot

positive association as x increases, y tends to increase; the points rise from left to right

negative association as x increases, y tends to decrease; the points fall from left to right

no association the points show no pattern; knowing x tells you nothing about y

linear association the points cluster around a straight line

nonlinear association the points follow a pattern that is not a straight line (often a curve)

Notes to Forgetful Self

Name	What the points do
positive linear	
negative linear	
no association	
nonlinear	
cluster	
outlier	

Must every point lie on one line for the association to be linear? Explain.